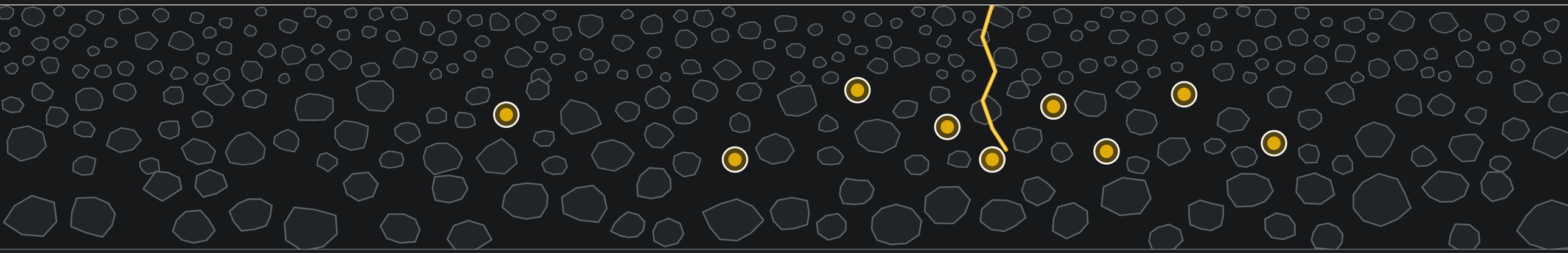
JKUAT TECH WEEK 6.0 / NETZERO CHALLENGE

# SMART ASPHALT

## Roads built from Kenya's waste that repair themselves using the sun.

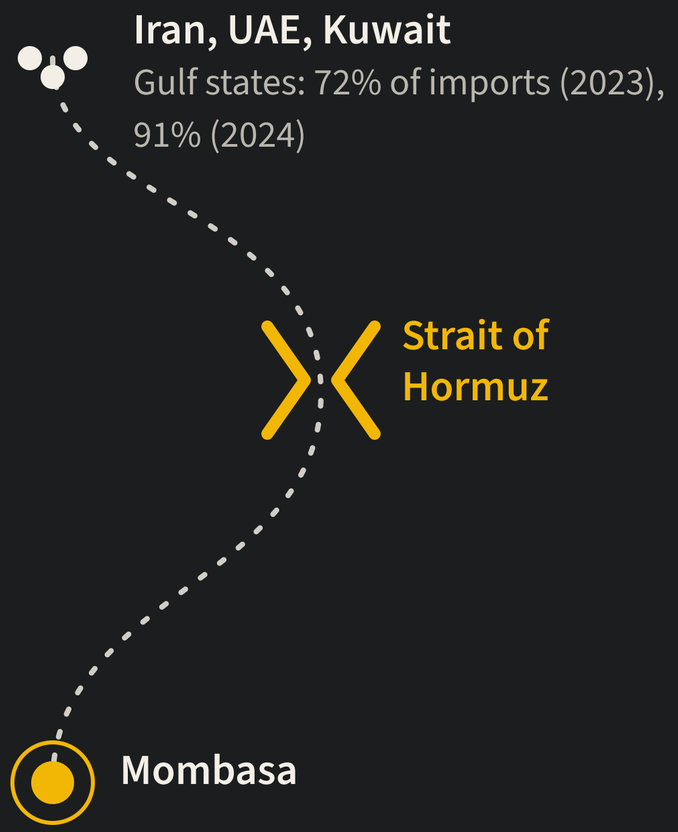


Andrew Weloba · Anthony Mugambi · Dennis Katiku · Gregory Miraya · Leslie Owuor · Trevor Mugo

# Kenya imports every gram of the bitumen in its roads.

# USD 20–28 million

a year, the recorded bitumen import bill (about KES 2.6–3.9 billion).




### WHAT IS BITUMEN?

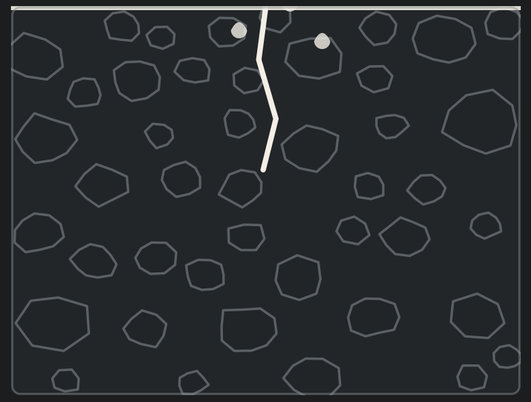
The black glue that holds a road’s stone together. A petroleum product, priced in dollars.

Source: UN Comtrade, Kenya imports of petroleum bitumen (HS 271320). 50,592 t worth USD 27.5 M in 2023; 39,850 t worth USD 20.1 M in 2024. Shares by weight.

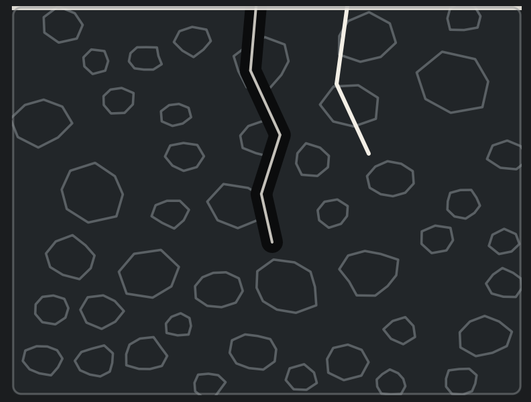
# And the roads still fail.



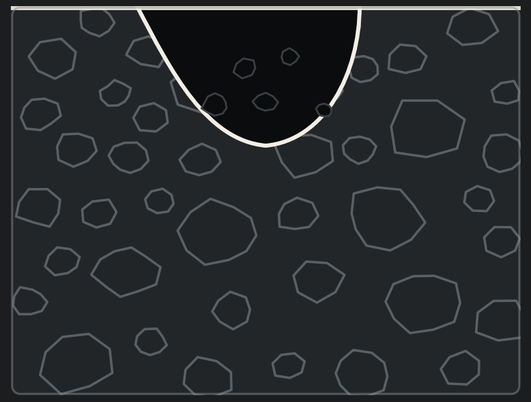
1 A hairline crack forms



2 Water gets in



3 The crack widens



4 A pothole opens

## 64°C

Kenya's maximum pavement design temperature. Hot binder softens.

## 15 → 5 years

A 15-year road can last about 5 under gross truck overloading.

## Brittle

Sun and oxygen slowly turn the binder stiff and easy to crack.

**In 2023, 30% of Kenya's roads were in poor condition and 46% in fair condition.**

Sources: Superpave temperature zoning for the Great Lakes region (Kenya max pavement design temperature 64°C); engineer quoted in The Standard (2015); Kenya Roads Board 2023 condition survey via Business Daily.

# Meanwhile, we throw away what could help.

| | |
|---|---|
| **0.5–1.3 million t** | of plastic waste a year. About 8% is recycled. |
| **3.2 million** | waste tyres a year (2024). About 11% are recycled. |
| **280,000 t** | of used cooking oil could arise a year. |
| **2.4 million t** | of sugarcane bagasse a year. About 1 million t is dumped or burned. |

## We import a fossil product to build roads, while the raw materials for an alternative go to waste.

Sources: Kenya Plastics Pact (via IPS, 2023); EcoPro/CSE (2024); Business Daily (700,000 t of oil used, about 40% becomes waste, only part is collected); Norges Vel. Most other bagasse fuels the mills.

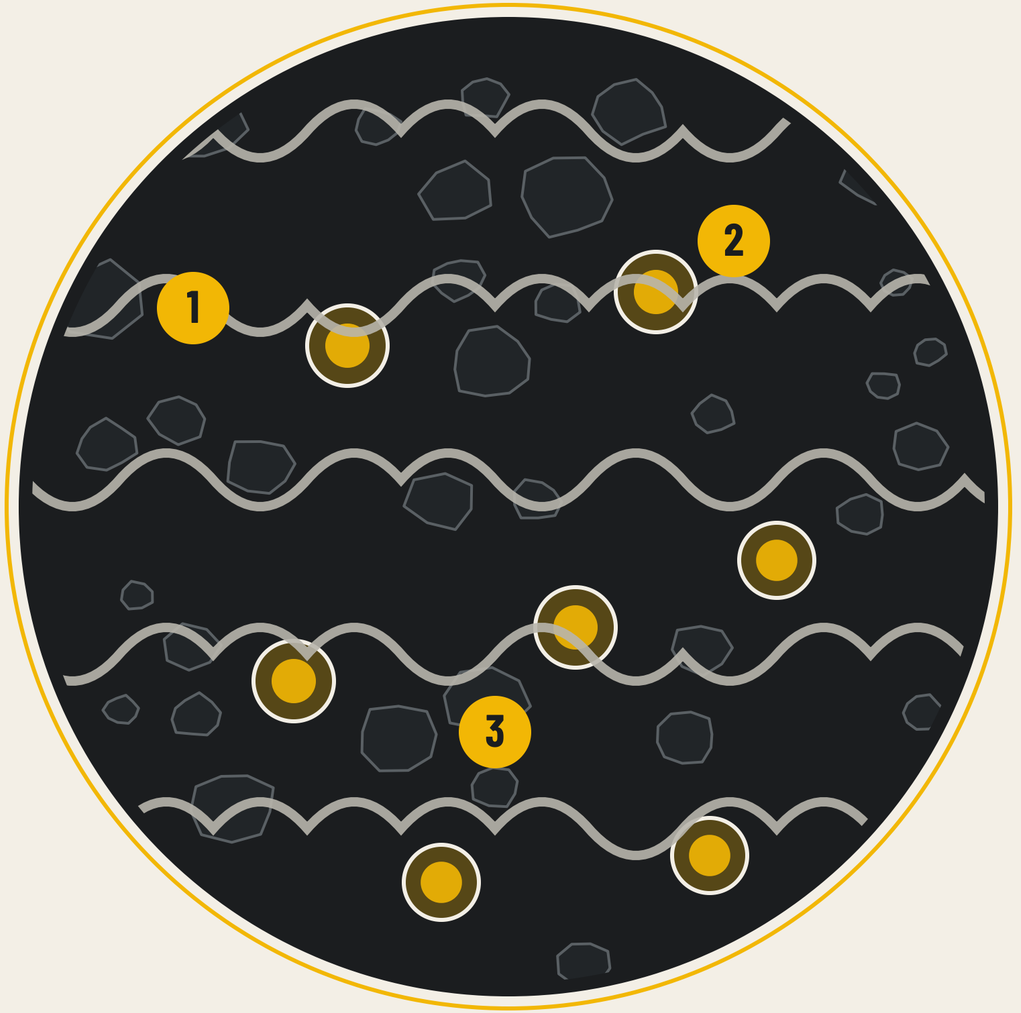
# The idea: a road that carries its own repair kit.

**1 Plastic backbone**
Recycled bags and bottle-cap plastic, melted in. A stiffer road that resists wheel ruts.

**2 Repair capsules**
Microscopic capsules of plant oil that open only where a crack forms.

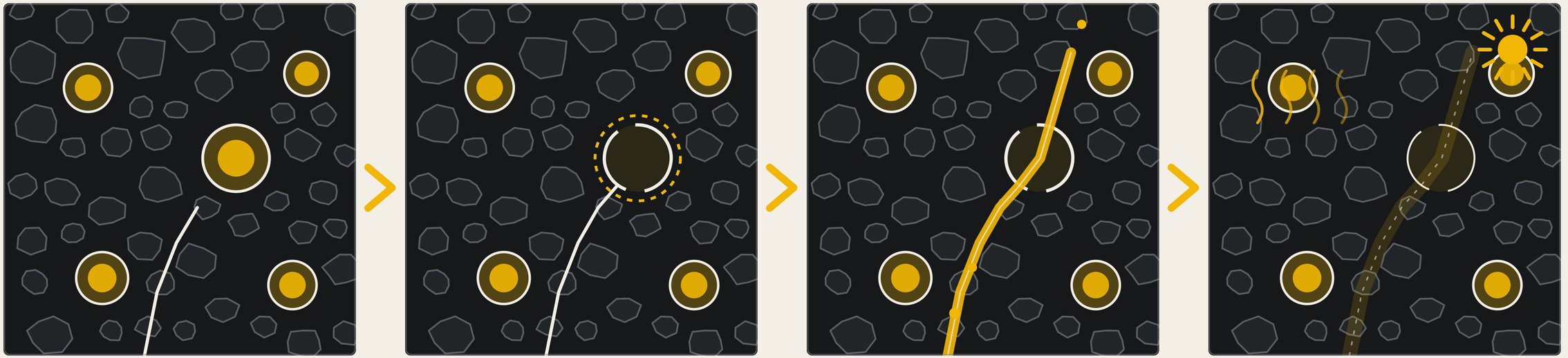
**3 Less imported bitumen**
10–15% less virgin bitumen in every tonne of mix (design target).

## Like skin healing a cut, the road repairs its own damage.




The binder under a microscope. Illustration, not to scale.

# How the road heals itself.



1. A crack forms under traffic and grows.
2. It reaches a capsule. The shell breaks.
3. Plant oil flows in and softens the aged binder.
4. The sun warms the road to 45–65°C. The crack faces flow together and bond.

**The design idea: capsules stay shut in everyday traffic and open at the tip of a crack, where stress is concentrated. Testing this is part of our lab plan.**

Illustration. Published lab healing results range from a few percent to about 80%, depending on the capsule, the test and the temperature. We have not measured it ourselves yet.

# Inside a capsule: plant oil in a seaweed shell.

**SHELL**

Calcium alginate from seaweed, reinforced with nano-silica so it survives hot asphalt mixing. About 2–4 µm thick.

**CORE**

Used cooking oil, castor oil and sugarcane-bagasse oil. It restores the oils that ageing takes out of bitumen.

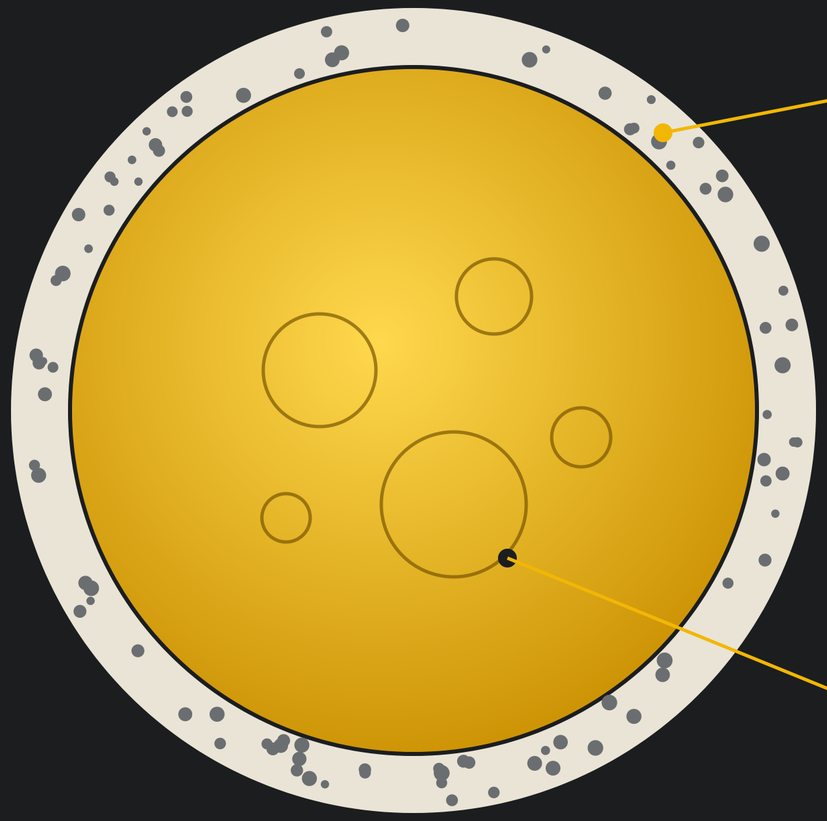



Illustration. Actual capsules are microscopic.

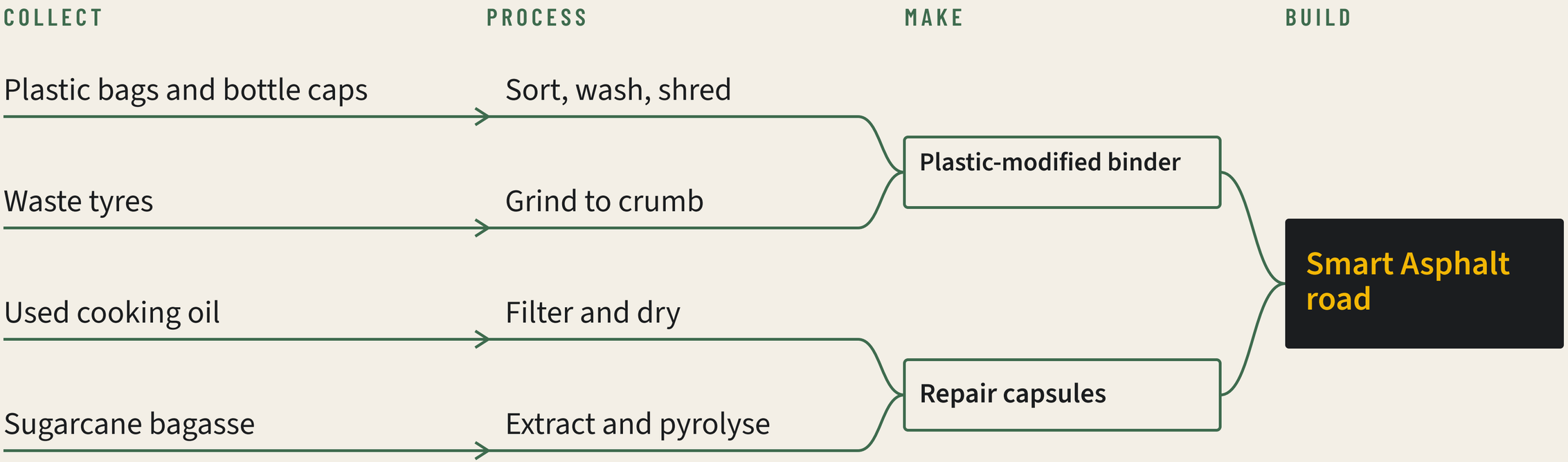


capsule 25–45 µm

grain of fine sand 60–200 µm

Our design. Published TU Delft capsules (calcium alginate, which survived asphalt production) are millimetre-sized; ours are a smaller design target. Capsules would make up 3–4% of the binder.

# From waste to road.



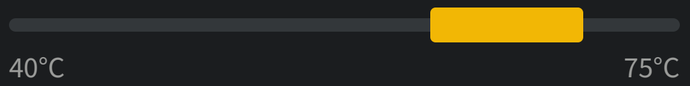

Used cooking oil is already collected in Nairobi for biodiesel and export. Roads are made in existing asphalt plants with standard paving equipment. Extra equipment costs about USD 0.4–0.8 million.

Sources: Phase 1, 3 and 4 studies. Crumb rubber is used in the Highland formulation.

# Built for Kenya’s three road conditions.

## Hot coast

Mombasa, Malindi, Lamu




**62–70°C** estimated hottest pavement

THE PROBLEM

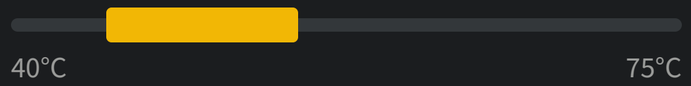
Extreme heat softens the road into wheel ruts.

OUR TWEAK

7% plastic and a thicker capsule shell.

## Nairobi highlands

Nairobi, Nyeri, Thika




**45–55°C** estimated hottest pavement

THE PROBLEM

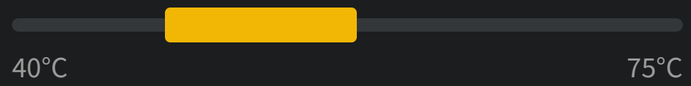
Heavy traffic causes fatigue cracking.

OUR TWEAK

4% capsules, plus crumb rubber from waste tyres.

## Black-cotton soil

Athi River, Konza, Kajiado




**48–58°C** estimated hottest pavement

THE PROBLEM

Clay swells and shrinks, tearing the road from below.

OUR TWEAK

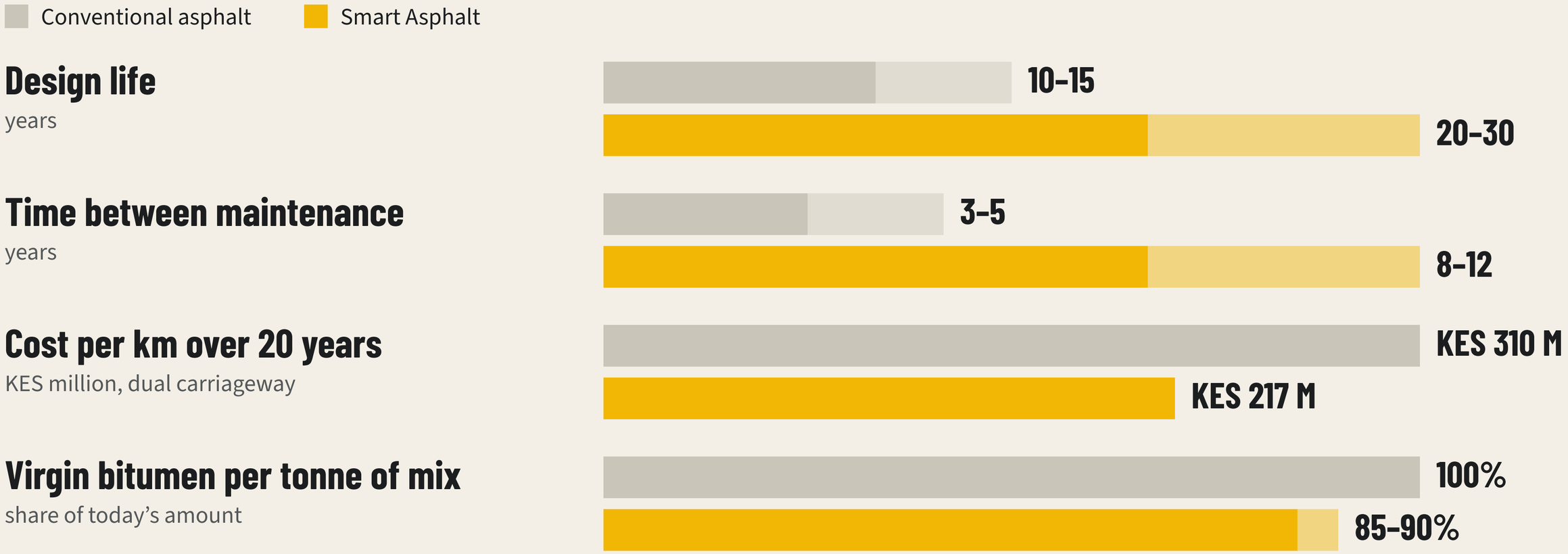
An elastic polymer, and the ground treated first.

## Kenya’s heat, usually a road’s enemy, is what drives the healing.

Zone temperatures are our estimates (Kenya's maximum pavement design temperature is 64°C). Black-cotton soil also needs ground treatment; no binder can stop ground movement alone.

# What we expect, as design targets.

NOT YET MEASURED




## All this for about 1.5% more to build.

Our own model, informed by published research. Lighter bars show the range. Not yet tested on a Kenyan road.

# Each part exists. The combination is new.

PROVEN ELSEWHERE

- ✓ India has built about 40,000 km of rural road and 2,900 km of highway with waste plastic.
- ✓ Plant-oil rejuvenators for old asphalt are sold commercially, for example Cargill's Anova.
- ✓ Calcium-alginate self-healing capsules are published lab research (TU Delft and others). Results vary widely.

WHAT IS NEW

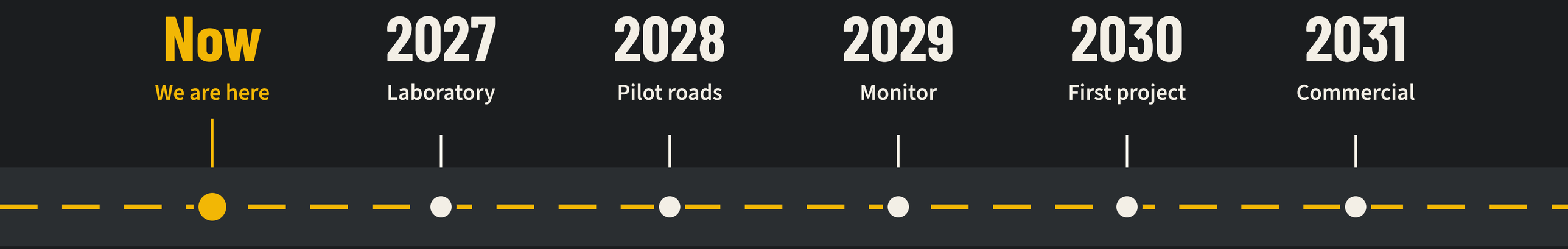
## All three in one binder, made from Kenyan waste and tuned to Kenyan heat and traffic.

**Stage:** laboratory validation (TRL 4–5). No Kenyan road trial yet.

| WHAT COULD GO WRONG | OUR ANSWER |
|---|---|
| Plastic separates from the bitumen in storage. | **A compatibiliser additive, and storage-stability tests on every batch.** |
| Healing may be small, or capsules may open under heavy traffic. | **Lab tests before any road. Tune the shell and the dose.** |
| Regulators and the public worry about plastic shedding. | **A monitored pilot road and a KEBS/KeNHA standard.** |

Sources: IDEAS Plastic Roads India (2025); Cargill; Xu et al. (2019) and TU Delft; Penn State and TRID on microplastics. See the full source ledger on the project website.

# From laboratory to pilot road in two years.




## The next step is small and specific: one lab, one road.

Source: Phase 1 and 4 implementation roadmaps. Timeline assumes lab access and a road-agency partner.

# One plant is a start. A standard is the goal.

## 80–140 km

of road a year from one plant, using 240–420 tonnes of plastic (our model).

That is under 0.1% of Kenya's plastic waste, so the aim is a national standard, not one factory.

HOW IT SCALES

- KeNHA, KURA and KeRRA road tenders
- Licences to asphalt producers
- East Africa

ENVIRONMENT

Plastic, tyres, cooking oil and bagasse kept out of dumpsites and drains. Less fossil bitumen, fewer rebuilds.

ECONOMY

Less foreign exchange spent on bitumen. A modelled saving of KES 93 million per km of dual carriageway over 20 years.

COMMUNITY

Income for waste collectors, sugar-belt communities and castor farmers. Safer, cheaper journeys.

**Circular Economy**
Primary

**Clean Energy**
Fossil use and sun-driven repair

**Energy Efficiency**
Warm-mix asphalt, fewer rebuilds

**Connected Technologies**
Pilot-road sensors

# Six civil engineering students, four phases of research.

BSC CIVIL ENGINEERING / JKUAT

## Andrew Weloba

## Anthony Mugambi

## Dennis Katiku

## Gregory Miraya

## Leslie Owuor

## Trevor Mugo

WHAT WE HAVE STUDIED

Capsule design for Kenyan conditions

The economics of Kenya’s bitumen imports

Ten alternative binder technologies

A full Smart Asphalt formulation

WHAT WE ARE LOOKING FOR

**A materials or chemistry lab partner.**

**A road authority to host a 500 m pilot.**

Study reports: Phase 1 (microcapsules), Phase 2 (macro-economics), Phase 3 (binder ladder), Phase 4 (Smart Asphalt design).

# The technology exists.
# The waste exists.
# What we need is a 500 metre road.

